$T(n) = 4T(n/2) + n \Rightarrow T(n) \in ?$

$T(n) = 4T(n/2) + n^2 \Rightarrow T(n) \in ?$

$T(n) = 4T(n/2) + n^3 \Rightarrow T(n) \in ?$

If $a < b^d$, $T(n) \in \Theta(n^d)$

If $a = b^d$, $T(n) \in \Theta(n^d \log n)$

If $a > b^d$, $T(n) \in \Theta(n^{\log_b a})$

$a=4, b=2, d=1$

$a > b^d$

$T(n) \in \Theta(n^{\log_2 4})$

$T(n) \in \Theta(n^2)$

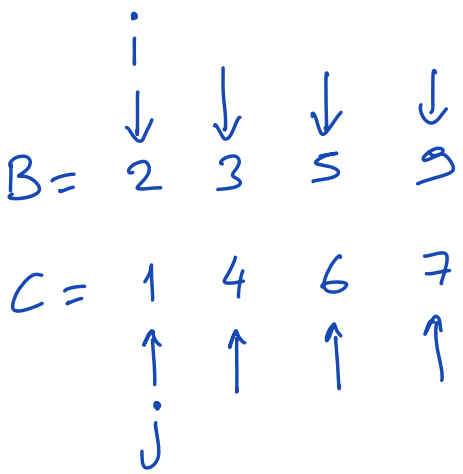
$a=4 \quad b=2 \quad d=2$

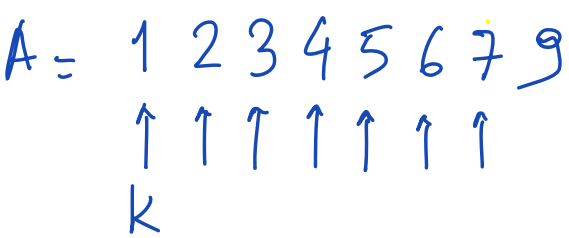
$a = b^d$

$T(n) \in \Theta(n^2 \log n)$

$a=4 \quad b=2 \quad d=3$

$a < b^d$ $T(n) = \in \Theta(n^3)$





Merge operation:

Best case:

B: 1 2 3 5 6 7 15.

C: 16 17 18 . . .

smallest element of C => > largest element of B.

$\frac{n}{2}$ comparisons.

Worst case

B: } last i & j increases until the end of the arr.
C: }

$n-1$ comparisons.

$$T(n) = 2T\left(\frac{n}{2}\right) + f(n)$$

↳ complexity of Merge op.

In the best case $2T\left(\frac{n}{2}\right) + \frac{n}{2}$
// worst case $2T\left(\frac{n}{2}\right) + n-1$
} $f(n) \in \Theta(n)$

$a=2$ $b=2$ $d=1$ $T(n) \in \Theta(n \log n)$.

ALGORITHM MergeSort3(A[0..n-1]) 3-way Mergesort.

if $n > 1$

copy $A[0..\lfloor n/3 \rfloor - 1]$ to B.

copy $A[\lfloor \frac{n}{3} \rfloor .. 2\lfloor \frac{n}{3} \rfloor - 1]$ to C

copy $A[2\lfloor \frac{n}{3} \rfloor .. n-1]$ to D

Mergesort3 (B)

Mergesort3 (C)

Mergesort3 (D)

Merge (B, C, E) → $\frac{2n}{3} - 1$

Merge (E, D, A) → $n - 1$

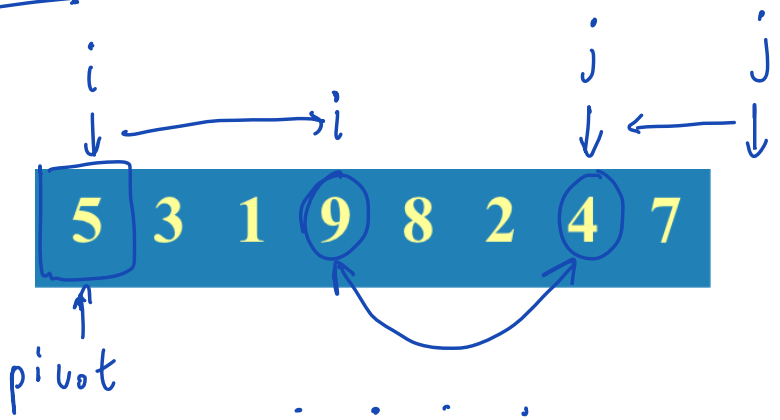
} $\frac{5n}{3} - 1$

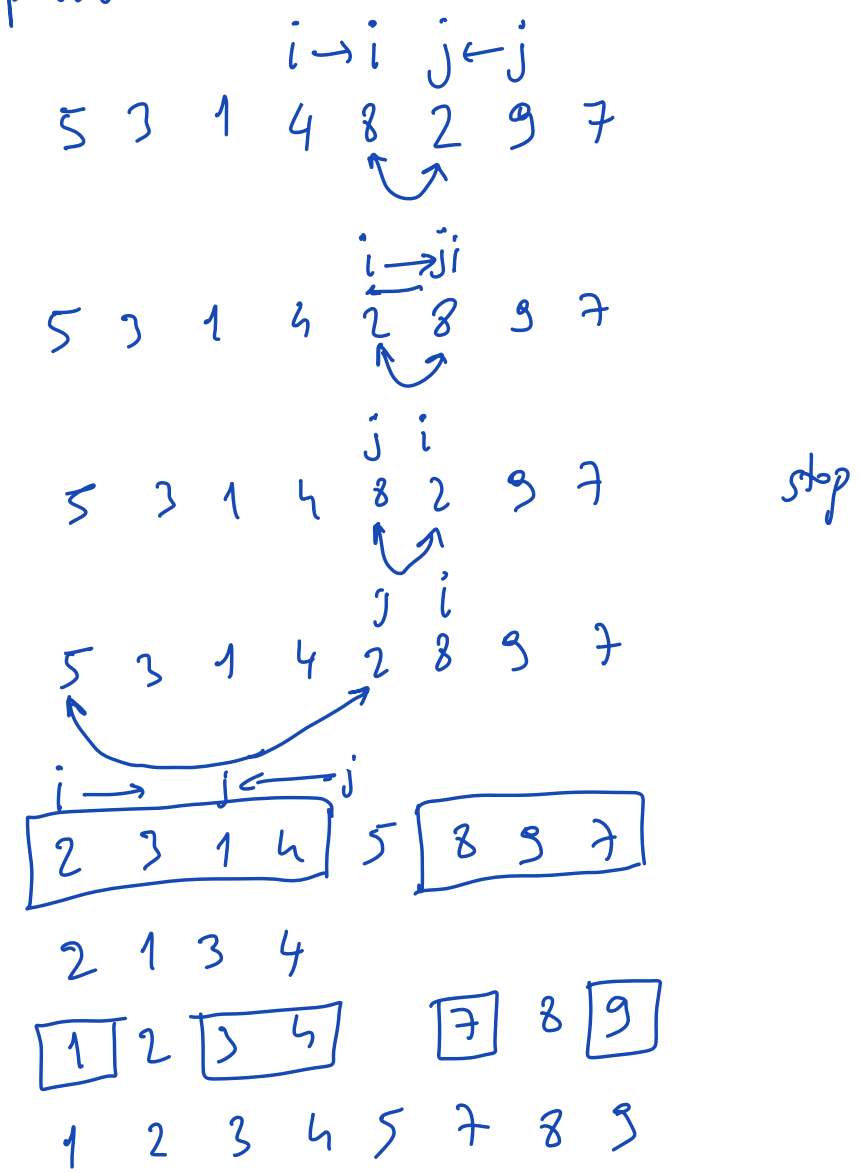
Worst case time complexity:

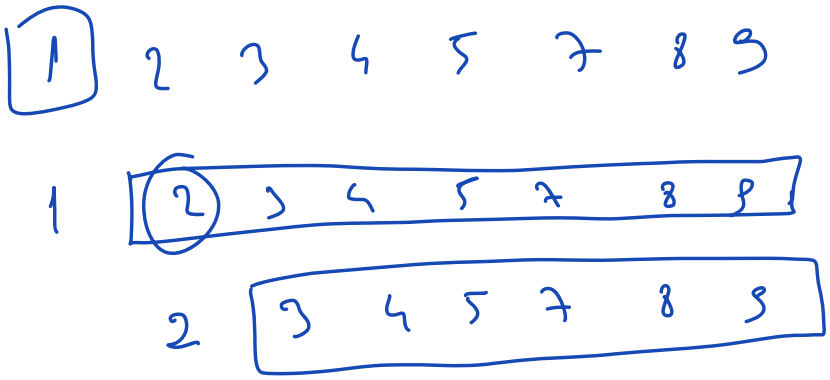
$$T(n) = 3T\left(\frac{n}{3}\right) + \underbrace{\left(\frac{5n}{3} - 1\right)}_{\in \Theta(n)}$$

$$T(n) \in \Theta(n \log n)$$

Quicksort:


i
i
j
j
5 3 1 9 8 2 4 7
pivot


i→i j←j
5 3 1 4 8 2 9 7
i→ji
5 3 1 4 2 8 9 7
j i
5 3 1 4 8 2 9 7
stop
j i
5 3 1 4 2 8 9 7
i→ j←j
2 3 1 4 5 8 9 7
2 1 3 4
1 2 3 4 5 7 8 9



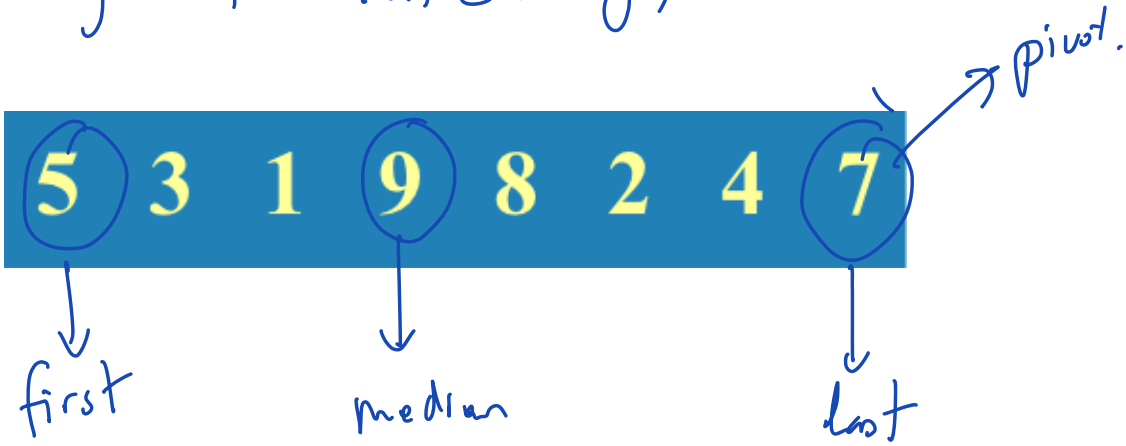
Worst case : $T(n) = T(n-1) + f(n)$ where $f(n) \in \Theta(n)$

$$T(n) \in \Theta(n^2)$$

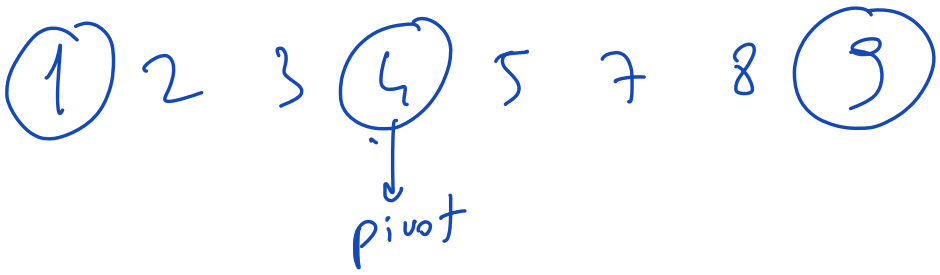
best case. $T(n) = 2T\left(\frac{n}{2}\right) + f(n)$ where $f(n) \in \Theta(n)$

$$T(n) \in \Theta(n \log n).$$

Avg case: $T(n) \in \Theta(n \log n)$.



median (5, 9, 7) = 7



$A = A_1 A_2$ $\quad$ $B = B_1 B_2$

n digits $\quad$ $\frac{n}{2}$ digits.

$$A * B = A_1 * B_1 * 10^n + \left((A_1 + A_2) * (B_1 + B_2) - A_1 * B_1 - A_2 * B_2\right) * 10^{\frac{n}{2}} + A_2 * B_2$$

multiplications decrease from 4 to 3.

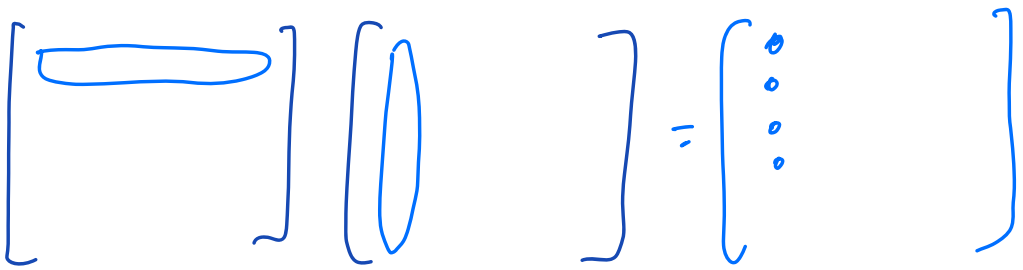
$M(n) = 3\, M\left(\frac{n}{2}\right)$ $\quad$ $a = 3 \quad b = 2 \quad d = 0$

$a > b^d \quad T(n) \in \Theta(n^{\log_2 3})$

$\Theta(n^{1.585})$

$\downarrow$

one-digit multiplications when the inputs are n digit.



n^2 entries.

for each n multiplications are required in brute-force method.

Brute-force $\rightarrow T(n) \in \Theta(n^3)$.

